Data Engineering

BY THE END:

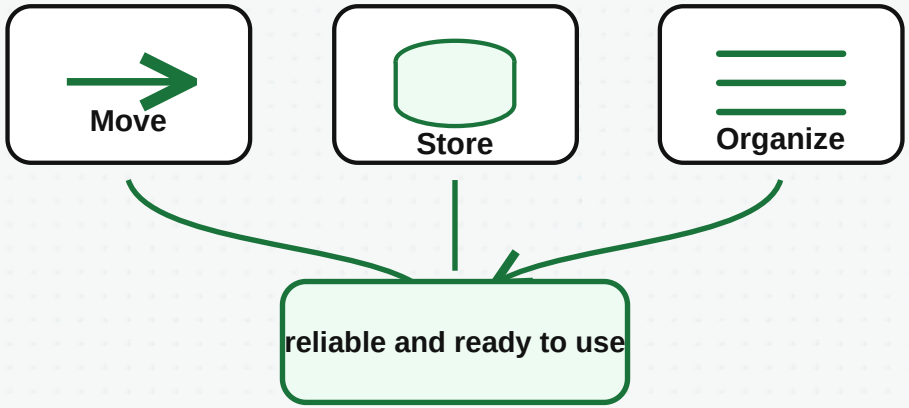
You can describe what data engineers build and how they prepare data for others to use.

Overview

An analyst opens a report and finds three different versions of the same sales numbers: one old, one incomplete, one in the wrong format. The data exists, but nobody can trust it yet. Before analysts and data scientists can work with data, it must be collected, stored, and prepared in a way people can trust. That is the role of **data engineering.**

What is data engineering?

Data engineering focuses on building and looking after the systems that move, store, and organize data so it is reliable and ready to use.



What data engineers do

Common responsibilities include:

- Building **data pipelines** that collect data from different sources
- Storing data in databases, data lakes, or a **data warehouse**
- Cleaning and changing large amounts of data
- Making sure data is available, safe, and easy to use
- Working with cloud tools and automation

If data analysis is about finding useful answers, data engineering is about making those answers possible.

Example

The wholesaler from lesson 2 has data arriving from four places at once:

- A till in each shop, recording sales as they happen
- Delivery drivers logging what was dropped off, and where
- Supplier invoices, arriving as PDFs by email
- Phone orders, written into a notebook and typed up at the end of the day

None of these agree on how a product is named, and two of them arrive late. A data engineer builds the pipelines that pull all four together, match the product names, and deliver one clean table that an analyst can actually use every morning, without someone doing it by hand every day.

